UNIT 3 · STANDARD 6.AT.3C

Ratio Reasoning: Convert Measurements within the Same System

Lesson 3-6

Name: _____ **Date:** _____ **Class:** _____

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can use ratio reasoning to convert between units within the same measurement system.

Language Objective: I can explain a conversion using the words unit ratio, convert, measurement system, and per.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Conversion factor | **Unit ratio** | **Convert** | **Measurement system** | **Double number line** | **Ratio table** | **Customary units** | **Metric units**

Level 3: try the blanks from memory first, then check the bank.

1 The ratio that trades one unit for another is a _____ _____.

2 A ratio comparing an amount to exactly _____ unit is a unit ratio.

3. To change a measurement to another unit without changing the amount is to _____.

4. Inches and feet belong to the customary _____; centimeters and meters to the metric one.

5. Two lines whose matching ticks show the same amount in two units form a _____ _____ _____.

6. A table whose every column holds the same comparison is a _____ _____.

7. Inches, feet, pounds, and gallons are _____.

8. Meters, grams, and liters are _____.

Write About the Math

How many cup-scoops of broth does the recipe need, and how do you know?
¿Cuántas tazas de caldo necesita la receta y cómo lo sabes?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Conversion factor** — *Factor de conversión*
- ☐ **Unit ratio** — *Razón unitaria*
- ☐ **Convert** — *Convertir*
- ☐ **Measurement system** — *Sistema de medidas*
- ☐ **Double number line** — *Recta numérica doble*

Model: *Find the scale factor (120 divided by 8 = 15), then multiply each ingredient by it.* — Students find the scale factor of 15 and explain that multiplying both ingredients by 15 keeps the recipe proportional (5 cups tomatoes becomes 75, 3 cups mozzarella becomes 45).

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The unit ratio is ___ cups to 1 quart. Cups are ___ than quarts, so I ___ to convert. 3 × 4 = ___ cups. The amount of broth did not change — only the ___ did.**
- **My answer is ___ because ___.**
  Mi respuesta es ___ porque ___.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ___.**
  Mi afirmación es ___.
- **My evidence is ___, so ___.**
  Mi evidencia es ___, entonces ___.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.